

Stephen Jonathan Pratama

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I am a high-achieving IT & Big Data Analytics professional specializing in architecting scalable backend systems, high-performance Agentic AI workflows, and LLM integrations. With a strong foundation in Python (FastAPI) and Node.js, I deliver enterprise-ready AI applications by leveraging prompt engineering, data preprocessing, and model fine-tuning. I am passionate about utilizing cloud-native technologies and vector databases to drive digital transformation, translating complex business requirements into robust technical solutions.

TECHNICAL SKILLS

- **Languages & Backend:** Python (FastAPI/Flask), Node.js (Express), PHP, C++, JavaScript.
- **AI & Machine Learning:** PyTorch, Hugging Face, LangChain, LangGraph, RAG, Prompt Engineering, Model Fine-Tuning, Google Gemini API, Computer Vision.
- **Databases & Tools:** PostgreSQL, MySQL, QdrantDB (Vector DB), Git, Redis, MinIO.
- **Cloud & DevOps:** AWS (EC2), Docker, GitLab CI/CD, Kubernetes, Linux.
- **Real-time & Security:** Apache Kafka, WebSockets, JWT, OAuth, SSH Encryption.

EDUCATION

CALVIN INSTITUTE OF TECHNOLOGY

Jakarta, Indonesia

Eight-semester student, IT & Big Data Analytics

2022-2026

GPA: 3.72 / 4

PROFESSIONAL EXPERIENCE

PT DIGDAYA SOLUSI TEKNOLOGI

Jakarta, Indonesia

Junior Full-Stack Developer

2025-Present

- Architected and optimized high-concurrency backend APIs using Node.js and FastAPI, improving data retrieval efficiency by 30% through advanced indexing and query optimization.
- Integrated Third-Party AI APIs (e.g., Google Gemini) and LLM functionalities, expanding platform capabilities while conducting prompt engineering to optimize AI model outputs.
- Improved software quality by implementing rigorous code reviews, automated testing, debugging, and maintaining technical documentation for AI and backend functionalities.
- Engineered a real-time vehicle tracking module by integrating Apache Kafka for high-throughput data streaming and WebSockets, reducing latency for live updates to under 100ms.
- Collaborated tightly with cross-functional teams (Product, Design, and Engineering) to translate complex business requirements into scalable technical architectures, ensuring 100% on-time delivery of product milestones.

RELATED PROJECTS

OBJECT STORAGE & CONTEXTUAL AI CHATBOT FOR EDUCATION

Jakarta, Indonesia

PT Digdaya Solusi Teknologi - Junior Software Developer

2026

- Architected a sophisticated Agentic AI system using LangGraph and LangChain to orchestrate complex workflows, enabling autonomous tool-usage such as automated email reporting via SMTP.
- Engineered a RAG (Retrieval-Augmented Generation) pipeline utilizing QdrantDB, reducing semantic search latency by 40% through custom HNSW indexing.
- Developed a Multimodal RAG Pipeline integrating Qwen 3-VL (via Ollama) and MinIO for S3-compatible storage, allowing the agent to "see" and reason over visual data in educational documents.
- Implemented a Hybrid Database Strategy combining PostgreSQL for relational metadata and Qdrant for high-dimensional embeddings, ensuring data consistency and retrieval accuracy.

SECURE ONLINE VIDEO CALL SYSTEM FOR INDONESIAN PRISONS

Jakarta, Indonesia

PT Digdaya Solusi Teknologi - Junior Software Developer

2025-Present

- Developed SILA (Sistem Layanan Audio Visual / Wartelsuspas), a secure online communication platform designed to enable encrypted video calls between Warga Binaan Lapas (inmates) with their families through encrypted video calls and virtual meetings.
- Built the backend system using Node.js (Express) and PostgreSQL, implementing robust authentication, session management, and data encryption.
- Implemented Google Gemini API for automatic meeting transcription and summarization, providing accessible records of conversations while maintaining strict confidentiality.
- Integrated FaceAPI.js for real-time facial verification to authenticate users during login, enhancing system security and preventing impersonation.
- Utilized WebSocket for real-time video call signaling and communication, enabling low-latency, reliable connections between users.
- Used FFmpeg to convert and process recorded meeting audio for storage, playback, and AI-based summarization pipelines.

VOLUNTEERING AT SABDA BIBLE FOUNDATION: BIBLE LEARNING PLATFORM

Jakarta, Indonesia

Volunteer - Full-Stack Developer & AI Integrator

2024

- Architected an end-to-end digital learning platform utilizing PHP, MySQL, and the Smarty template engine, managing the full lifecycle from backend logic to responsive frontend UI integration.
- Engineered an automated data extraction pipeline utilizing Python to scrape unstructured audiovisual content from YouTube for large-scale content processing.
- Integrated OpenAI's Whisper and GPT APIs to perform automated speech-to-text transcription and advanced NLP tasks, generating structured summaries, categorizations, and interactive discussion prompts.
- Implemented Elasticsearch to ensure lightning-fast content retrieval and advanced filtering capabilities for users across the platform.

FINETUNE A DEEPSEEK R1 TRANSFORMER MODEL FOR TEXT SUMMARIZATION

Jakarta, Indonesia

Student - Junior AI Engineer

2025

- Developed a text summarization system by fine-tuning the DeepSeek-R1 Distill Llama-8B model using Python, PyTorch, and Hugging Face Transformers.

- Prepared and preprocessed diverse datasets, executing text cleaning, tokenization, data labeling, and handling domain-specific terminology for model evaluation.
- Applied transfer learning with LoRA (Low-Rank Adaptation) and experimented with mixed-precision training (FP16) to optimize inference latency without sacrificing accuracy.